

How to check for overlapping intervals

The easiest overlap test starts by asking when overlap is impossible. [\[1\]](#)



1. Adapted from Zayenz, "How to check for overlapping intervals," zayenz.se, 2025.

Use half-open intervals

For this talk, an interval is written as

$[s, e)$ with $s < e$.

The start belongs to the interval. The end does not.

○ **CONTAINS 3**

$[1, 4)$

□ **DOES NOT CONTAIN 4**

$[1, 4)$

Listing overlap cases gets awkward

○ ENDPOINTS CROSS

- A starts first
- B starts first

□ ONE CONTAINS THE OTHER

- A contains B
- B contains A

Four valid cases are already too many for a memorable test.

Separation has only two cases

Two intervals do not overlap when one ends before the other starts.

1. A is before B: $A.end \leq B.start$
2. B is before A: $B.end \leq A.start$

Negate separation

NO OVERLAP

$$(a_e \leq b_s) \vee (b_e \leq a_s)$$

1 NEGATE

$$\neg((a_e \leq b_s) \vee (b_e \leq a_s))$$

2 APPLY DE MORGAN

$$(b_s < a_e) \wedge (a_s < b_e)$$

The implementation follows the derivation

```
@dataclass
class Interval:
    start: int
    end: int
    def overlaps(self, other: "Interval") -> bool:
        return (
            other.start < self.end
            and self.start < other.end
        )
```

The two comparisons say that each interval starts before the other one ends.

Boundary cases follow from the notation

○ ADJACENT

$[1, 4)$ $[4, 7)$

$4 \leq 4 \implies$ no overlap

□ SHARED RANGE

$[1, 5)$ $[4, 7)$

$4 < 5 \wedge 1 < 7 \implies$ overlap

Detail: closed intervals change the boundary

For closed intervals $[s, e]$, touching at one endpoint counts as overlap.

The corresponding comparisons use $\leq$ rather than $<$.

Boxes repeat the test on each axis

Two axis-aligned boxes overlap when their horizontal intervals overlap and their vertical intervals overlap.

$$\begin{array}{l} b_l < a_r \quad \wedge \quad a_l < b_r \\ b_b < a_t \quad \wedge \quad a_b < b_t \end{array}$$

The interval argument avoids sixteen geometric cases.

Check the complement first

When a Boolean property has many positive cases, its negation may have a much smaller case analysis.

For interval overlap:

$$b_s < a_e \wedge a_s < b_e$$